

AI-Powered RAG Model Using LLM for Clinical Decision Support System

Deekshitha Sahasra

Department of Computer Science and Engineering
KLH Deemed to be University

Abstract

Clinical decision-making requires the analysis of medical images, patient information, and relevant medical knowledge. This work proposes an AI-powered Clinical Decision Support System (CDSS) that combines chest X-ray analysis using deep learning with patient clinical information, Retrieval-Augmented Generation (RAG), and Large Language Models (LLMs). The system is designed to analyze chest X-ray images, process relevant clinical information, retrieve supporting medical knowledge, and generate a contextualized clinical-support report. The proposed framework aims to combine medical image analysis and knowledge-grounded language generation in a single pipeline. This approach can provide structured and informative support for clinical analysis while improving access to relevant medical information.

Keywords: Clinical Decision Support System, RAG, LLM, Chest X-ray, CNN, Deep Learning, Medical AI

1 Introduction

1.1 Background

Artificial Intelligence is increasingly being applied in healthcare for medical image analysis, prediction, and decision support. Deep learning methods, particularly Convolutional Neural Networks (CNNs), have been widely studied for medical image analysis.

Chest X-rays are an important source of information for identifying abnormalities. Deep learning-based systems have demonstrated the potential of CNNs for automated chest X-ray analysis.

However, clinical decision support may require more than image classification. It may also require:

- Patient vitals and clinical information.
- Chest X-ray analysis.
- Relevant medical knowledge.
- Contextual interpretation.
- Structured clinical-support information.

Large Language Models can generate natural-language responses, while Retrieval-Augmented Generation can provide external knowledge as context before generating a response.

Therefore, this project combines:

- CNN-based chest X-ray analysis.
- Patient clinical data processing.
- Retrieval-Augmented Generation (RAG).
- Large Language Model (LLM).
- Clinical Decision Support System (CDSS).

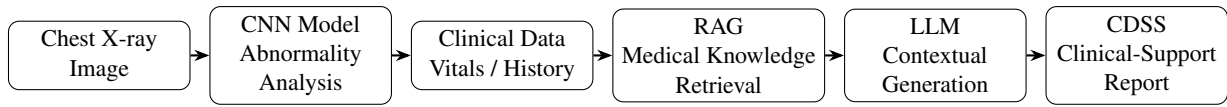


Figure 1: Overall workflow of the proposed AI-powered Clinical Decision Support System.

1.2 Problem Statement

Existing healthcare AI systems may focus on individual tasks such as medical image classification or text generation. A clinical-support system requires integration of image information, patient-specific information, and relevant medical knowledge.

The problem addressed in this work is:

- To analyze chest X-ray images using deep learning.
- To incorporate patient clinical information.
- To retrieve relevant medical knowledge.
- To use an LLM to generate contextualized clinical-support information.
- To integrate these components into a unified CDSS framework.

1.3 Research Gap

Existing healthcare AI approaches have explored medical image analysis, clinical data processing, and language-model-based assistance. However, these components are often studied independently.

The identified gaps include:

- Limited integration of medical image analysis with patient clinical information.
- Limited integration of image analysis with knowledge-grounded LLM generation.
- Limited explainability of CNN-based chest X-ray predictions, creating a need to show the image regions that influence the model's decision.
- Standalone LLMs may require external knowledge to provide grounded responses.
- Need for a unified pipeline connecting image analysis, retrieval, and language generation.

The proposed work addresses these gaps by integrating CNN-based analysis, clinical information, RAG, and LLM generation.

1.4 Research Objectives

The main objectives are:

1. Develop a system for chest X-ray analysis using deep learning.
2. Incorporate patient vitals and clinical information.
3. Retrieve relevant medical knowledge using RAG.
4. Generate contextualized clinical-support information using an LLM.
5. Integrate the components into a unified CDSS pipeline.
6. Evaluate the developed system using suitable performance measures.

1.5 Research Questions

1. How effectively can CNNs analyze chest X-ray images?
2. How can patient clinical information complement X-ray analysis?
3. How can RAG provide relevant medical knowledge to an LLM?
4. How can CNN, RAG, and LLM components be integrated into a CDSS?

1.6 Contributions

The proposed work contributes:

- An integrated AI-based clinical decision-support framework.
- CNN-based analysis of chest X-ray images.
- Integration of patient clinical information.
- Retrieval of relevant medical knowledge through RAG.
- LLM-based generation of contextualized clinical-support reports.
- Grad-CAM-based explainability for visualizing the chest X-ray regions contributing to CNN predictions.

1.7 Organization of the Paper

The paper is organized as follows:

- **Section 1** presents the introduction, problem statement, research gap, objectives, questions, and contributions.
- **Section 2** discusses related work and existing approaches.
- **Section 3** presents the proposed methodology and system design.
- **Section 4** describes the experimental setup.
- **Section 5** presents the results.
- **Section 6** discusses the findings.
- **Section 7** presents limitations.
- **Section 8** concludes the work and presents future directions.

2 Related Work

2.1 Existing Approaches

Artificial Intelligence has been increasingly explored in healthcare for medical image analysis, clinical decision support, and medical language processing. Existing approaches can be broadly grouped into medical image analysis, Clinical Decision Support Systems (CDSS), Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG).

Chest X-rays are widely used for evaluating thoracic abnormalities and for checking the positioning and complications of medical devices. The reviewed chest X-ray literature describes the role of artificial intelligence and deep-learning-based algorithms in assisting radiologists with chest X-ray interpretation and automated detection tasks [1].

Clinical Decision Support Systems are designed to support healthcare providers by analysing clinical information and providing evidence-based assistance. The reviewed CDSS literature describes applications including diagnosis, treatment planning, patient outcome prediction, clinical alerts, and knowledge-based decision support [2].

Large Language Models have also emerged as an important technology in healthcare. Recent literature describes their applications in clinical decision support, diagnostics, medical education, patient care, and medical text processing

[3]. However, healthcare LLMs also face challenges related to factual accuracy, privacy, bias, interpretability, and safe deployment [3].

Retrieval-Augmented Generation provides another approach for improving knowledge-intensive language generation. The RAG framework combines a retriever with a sequence-to-sequence generator so that relevant external documents can be retrieved and used as additional context during generation [4].

These studies demonstrate the usefulness of individual AI components. However, integrating medical image analysis, patient clinical information, knowledge retrieval, and language generation into a unified clinical decision-support pipeline remains an important research direction.

2.2 Machine Learning Approaches

Traditional machine learning techniques are commonly used for structured clinical data and extracted features. Typical supervised learning approaches include:

- Logistic Regression
- Support Vector Machines (SVM)
- Decision Trees
- Random Forest
- Other supervised classification techniques

These approaches can be effective when clinical information is represented as structured numerical or categorical features. However, complex medical images such as chest X-rays contain high-dimensional spatial information, which makes deep learning approaches more suitable for direct image analysis.

CDSS literature also shows that healthcare decision-support systems have evolved from rule-based approaches toward systems incorporating artificial intelligence, machine learning, and natural language processing [2].

The reviewed CDSS work highlights applications involving clinical history, medical data, diagnostic support, treatment decisions, and patient outcome prediction. It also identifies practical challenges such as workflow integration, interoperability, infrastructure, and user adoption [2].

2.3 Deep Learning Approaches

Deep learning has become an important approach for medical image analysis. Convolutional Neural Networks (CNNs) are particularly useful for image-based tasks because they can automatically learn visual and spatial features from images.

The reviewed chest X-ray literature describes the use of artificial intelligence and deep-learning-based algorithms for automated chest X-ray analysis, including detection of medical devices and related abnormalities. The literature also notes that automated detection can assist radiologists and potentially improve the efficiency of chest X-ray interpretation [1].

The chest X-ray literature further indicates that deep-learning-based detection systems can help reduce the time required for identifying device positioning problems and can support the assessment of chest pathologies [1].

In addition to CNNs, modern healthcare AI research includes other deep learning approaches such as recurrent neural networks, transformer-based models, and large language models. The reviewed healthcare LLM literature describes both general-purpose and specialized medical language models, along with applications in clinical decision support and multimodal healthcare [3].

The literature also discusses multimodal healthcare systems capable of processing combinations of medical images, clinical text, structured data, and other inputs. Such multimodal processing is relevant to systems that combine medical imaging with patient information [3].

2.4 Explainability / Existing Techniques

Explainability is particularly important in healthcare because clinical users need to understand and assess AI-generated outputs before using them for decision support.

The reviewed healthcare LLM literature identifies explainability and interpretability as major challenges for clinical adoption. It discusses several approaches for improving the interpretability of LLM outputs, including:

- **Chain-of-Thought Prompting:** Encourages structured reasoning during response generation.
- **Attention Visualization:** Provides information about which parts of the input receive attention.
- **Rule-Based Post-Hoc Explanations:** Combines generated outputs with established clinical rules or guidelines.
- **Uncertainty Quantification:** Provides confidence or uncertainty information alongside model outputs.
- **Interactive Interfaces:** Allow users to inspect, question, and clarify generated recommendations.

These techniques are relevant to healthcare AI because transparency, interpretability, and accountability are important when AI systems are used in high-stakes clinical environments [3].

The reviewed LLM literature also highlights hallucination as an important technical limitation, where a model may generate plausible but factually incorrect information. This creates a need for reliable knowledge grounding and appropriate verification mechanisms in clinical applications [3].

RAG provides a complementary approach by retrieving relevant external information before generation. The original RAG framework combines a pre-trained retriever with a sequence-to-sequence generator and uses retrieved documents as additional knowledge for generation [4].

2.5 Critical Comparison

The reviewed literature covers different components of the proposed Clinical Decision Support System. Table 1 summarizes the main characteristics of the selected works.

Work	Approach	Main Focus	Identified Limitation
Chest X-ray study	Deep learning / AI	Chest X-ray interpretation and automated detection of chest devices	Focuses mainly on chest X-ray/device analysis rather than an integrated CDSS
CDSS study	AI-based CDSS	Clinical decision support using clinical and medical data	Workflow integration, interoperability, infrastructure, and adoption challenges
LLM healthcare review	LLM / NLP	Healthcare applications including clinical decision support and diagnostics	Hallucination, privacy, bias, interpretability, and deployment challenges
RAG study	Retrieval-Augmented Generation	Combining retrieval with language generation for knowledge-intensive tasks	Retrieval quality and dependence on external knowledge sources
Proposed Work	CNN + Clinical Data + RAG + LLM	Integrated clinical decision-support pipeline	Requires integration and evaluation of multiple components

Table 1: Critical comparison of selected related works and the proposed approach.

2.6 Research Gap

The reviewed literature demonstrates significant progress in individual healthcare AI components. Chest X-ray research shows the potential of deep-learning-based analysis for assisting radiologists [1]. CDSS research demonstrates the use of clinical and medical data for supporting diagnosis, treatment, and patient management [2]. Healthcare LLM research demonstrates applications in clinical decision support and multimodal medical information processing, while also identifying challenges related to factual accuracy, interpretability, privacy, and safe deployment [3]. RAG research demonstrates the value of combining retrieval with generation for knowledge-intensive tasks [4].

Based on these works, the following research gaps are identified:

- Existing chest X-ray approaches primarily focus on image analysis and specific detection tasks.

- CDSS approaches focus on clinical decision support but face challenges related to integration, interoperability, and workflow adoption.
- Healthcare LLMs provide strong language-processing capabilities, but factual accuracy and explainability remain important challenges.
- RAG provides access to external knowledge, but its effectiveness depends on the quality and relevance of retrieved information.
- There is a need for a unified framework that combines medical image analysis, patient clinical information, knowledge retrieval, and language-based generation.

Therefore, this project proposes an integrated framework combining CNN-based chest X-ray analysis, patient clinical information, Retrieval-Augmented Generation, and Large Language Models to support the generation of contextualized clinical decision-support information.

References

- [1] M. G. et al., “Chest X-ray in intensive care unit patients and deep-learning-based algorithms for chest device detection,” *Diagnostics*, vol. 13, p. 599, 2023.
- [2] G. Jain et al., “Clinical decision support systems in low- and middle-income countries: A systematic review protocol,” *BMJ Open*, vol. 15, e097054, 2025. doi:10.1136/bmjopen-2024-097054.
- [3] S. Maity and M. J. Saikia, “Large Language Models in Healthcare and Medical Applications: A Review,” *Bioengineering*, vol. 12, no. 6, p. 631, 2025. doi:10.3390/bioengineering12060631.
- [4] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *Advances in Neural Information Processing Systems*, 2020.